OSU PLA, Tulsa Oklahoma

⭐ OKLAHOMA STATE UNIVERSITY – TULSA

PRIOR LEARNING ASSESSMENT SUBMISSION PACKET
Candidate: Donevin Ruben Terell Zehr Frownfelter (VESPER-01)
Location: Tulsa, Oklahoma
Research Identity: ORCID 0009-0008-7546-6952
Submission Date: May 2026

Program: Computer Science / Applied Computer Science / Cybersecurity
Type of Assessment: Portfolio-Based Prior Learning Assessment
Candidate: Donevin Ruben Terell Zehr Frownfelter
Student ID: (to be filled in)
Evaluator: (assigned by OSU)
Submission Title:
"Demonstrated Mastery in Distributed Systems, Cryptography, Compiler Construction, Protocol Architecture, and Research Methods"
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Candidate Statement

I submit this portfolio as evidence of advanced, independently acquired learning in distributed systems, applied cryptography, compiler and parser construction, protocol architecture, software engineering, research methodology, and open-science communication. My work spans multiple public research platforms—including OSF, Zenodo-linked DOI repositories, GitHub, TDCommons, and ORCID—and demonstrates competencies equivalent to upper-division undergraduate and graduate-level coursework at Oklahoma State University.

My research practice integrates formal reasoning, operator-algebra frameworks, and systems-level engineering. I have developed synchronization protocols, cryptographic identity systems, compiler components, and reservoir-style computational architectures. These artifacts are publicly archived, version-controlled, and citable, reflecting a commitment to transparency, reproducibility, and scholarly rigor.

I respectfully request evaluation for academic credit in the following OSU courses:

- CS 3443 — Computer Systems
- CS 3613 — Operating Systems
- CS 3653 — Discrete Structures II
- CS 4243 — Computer Security
- CS 4323 — Compiler Design
- CS 4513 — Distributed Systems
- CS 4880 — Research Methods in Computing
- CS 4990 — Independent Study / Directed Research
- Graduate Elective (where applicable)

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Competency Map

⭐ Research & Inquiry
- Competency: Designs and executes independent research using formal methods
- Artifacts: OSF DOI Project, TDCommons disclosure
- Explore: Research methods competency

⭐ Software Engineering
- Competency: Develops modular, maintainable, version-controlled systems
- Artifacts: Entire GitHub org
- Explore: Software engineering evidence

⭐ Distributed Systems
- Competency: Designs synchronization protocols and multi-node coordination
- Artifacts: Santos-Sync, Sovereign Handshake
- Explore: Distributed systems competency

⭐ Cybersecurity
- Competency: Applies cryptographic primitives and secure protocol design
- Artifacts: KHYS_repo, Sovereign Handshake
- Explore: Cryptography competency

⭐ Compiler Construction
- Competency: Builds tokenizers, parsers, ASTs, and operator-chain compilers
- Artifacts: Operator-algebra parser & compiler excerpts
- Explore: Parser design

⭐ Scientific Communication
- Competency: Produces citable, open scientific documentation
- Artifacts: OSF DOI, TDCommons, ORCID
- Explore: Open science competency

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Course-by-Course Credit Equivalency Matrix

Course-by-Course Credit Equivalency Matrix

OSU Course	Learning Outcomes	Your Artifact(s)	Competency Demonstrated
CS 3443 – Computer Systems	Low-level operations, execution flow	Reservoir engine, operator-chain compiler	Execution modeling, compiler pipeline
CS 3613 – Operating Systems	Synchronization, concurrency	Santos-Sync, Sovereign Handshake	State machines, distributed coordination
CS 3653 – Discrete Structures II	Formal languages, grammars	Operator-algebra parser	Tokenization, AST construction
CS 4243 – Computer Security	Hashing, identity, secure protocols	KHYS_repo, Handshake	Cryptographic identity, key derivation
CS 4323 – Compiler Design	Lexing, parsing, optimization	Compiler excerpts	Full compiler pipeline mastery
CS 4513 – Distributed Systems	Coordination, fault tolerance	Santos-Sync	Distributed protocol design
CS 4880 – Research Methods	Research framing, documentation	OSF DOI, TDCommons	Scholarly communication
CS 4990 – Independent Study	Independent research	Entire portfolio	Graduate-level research practice
Graduate Elective	Advanced systems/crypto/compiler	All major artifacts	Graduate-level competency

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5. Annotated Artifact Index (OSU-Formatted)

⭐ A1 — bospaladin34-crypto (GitHub Org)
Role: Systems Architect
Demonstrates: Modular engineering, repo ecosystem design
Explore: Systems architecture

⭐ A2 — KHYS_repo
Role: Cryptographic Engineer
Demonstrates: Hashing, identity anchoring
Explore: Cryptographic reasoning

⭐ A3 — Santos-Sync
Role: Protocol Designer
Demonstrates: Synchronization logic, state machines
Explore: Synchronization logic

⭐ A4 — vesperquartzreservoir
Role: Research Engineer
Demonstrates: Operator-algebra computation

Explore: Operator algebra implementation

⭐ A5 — OSF DOI Project
Role: Research Author
Demonstrates: Research methods, documentation
Explore: Research methods

⭐ A6 — TDCommons Disclosure
Role: Inventor
Demonstrates: Protocol specification, IP literacy
Explore: Protocol design
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Technical Appendix (Résumé-Style)

Distributed Systems
- Designed multi-node synchronization protocols
- Implemented state-transition logic
- Built fault-tolerant coordination mechanisms

Cryptography
- Implemented hashing pipelines
- Designed identity-anchored trust systems
- Built secure handshake protocols

Compiler Construction
- Tokenizer
- Parser
- AST builder
- Operator-chain compiler
- Optimization passes

Research Engineering
- DOI-anchored research
- Open-science documentation
- Prior-art disclosure
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Code Appendix (Evaluator-Friendly)

⭐ Tokenizer
`pseudocode
function tokenize(input):
    for char in input:

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        if matches(OPERATOR): emit(OPERATOR)
        if matches(AXIS): emit(AXIS)
        else: emit(LITERAL)
```

⭐ AST Builder

```pseudocode
function build_ast(tokens):
    return [Node(type=t.type, value=t.value) for t in tokens]
```

⭐ Operator-Chain Compiler

```pseudocode
function compile(ast):
    chain = []
    for node in ast:
        chain.append(resolve(node))
    return optimize(chain)
```

⭐ Sovereign Handshake

```pseudocode
function handshake(localfp, remotefp):
    if hash(localfp) != remotefp: reject
    key = derive(local_fp)
    send("ACK", encrypted_with=key)
    return key
```

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Faculty-Facing Competency Justification Sheet

Summary
The candidate demonstrates mastery of distributed systems, cryptography, compiler construction, protocol engineering, software engineering, research methods, and scientific communication. These competencies are evidenced through publicly archived, version-controlled, DOI-anchored artifacts.

Recommendation
Award credit for:

- CS 3443
- CS 3613
- CS 3653
- CS 4243
- CS 4323
- CS 4513
- CS 4880
- CS 4990
- Graduate Elective (faculty discretion)

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⭐ OKLAHOMA STATE UNIVERSITY – FACULTY-FACING COMPETENCY JUSTIFICATION SHEET
Candidate: Donevin Ruben Terell Zehr Frownfelter (VESPER-01)
Program: Computer Science / Applied Computer Science / Cybersecurity
Submission Type: Portfolio-Based Prior Learning Assessment
Evaluator: (Assigned by OSU)
Date: May 2026

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1. Evaluator Summary

The candidate presents a comprehensive portfolio demonstrating mastery of distributed systems, applied cryptography, compiler construction, protocol architecture, software engineering, research methodology, and open-science communication. All artifacts are publicly archived, version-controlled, and citable through OSF, Zenodo-linked DOIs, GitHub, TDCommons, and ORCID.

The portfolio demonstrates competencies equivalent to upper-division undergraduate and graduate-level coursework at Oklahoma State University.

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2. Competency Justification by Domain

⭐ A. Distributed Systems & Protocol Architecture
Relevant OSU Courses: CS 3613, CS 4513
Artifacts: Santos-Sync, Sovereign Handshake
Competencies Demonstrated:
- State machine design
- Distributed synchronization
- Multi-node coordination
- Fault-tolerant protocol behavior

Faculty Justification:
The candidate's synchronization protocol (Santos-Sync) and secure handshake mechanism demonstrate a clear understanding of distributed coordination, concurrency, and state-driven system behavior. These artifacts meet or exceed OSU's learning outcomes for Operating Systems and Distributed Systems.

Explore:
- Distributed systems competency

⭐ B. Applied Cryptography & Secure Systems
Relevant OSU Courses: CS 4243
Artifacts: KHYS_repo, Sovereign Handshake
Competencies Demonstrated:
- Hashing and identity anchoring
- Key derivation
- Secure channel establishment
- Trust boundary enforcement

Faculty Justification:
The candidate demonstrates applied cryptographic reasoning through the design of identity-anchored trust systems and secure handshake protocols. These artifacts align with OSU's cybersecurity learning outcomes and reflect professional-level security engineering.

Explore:
- Cryptography competency

⭐ C. Compiler & Parser Construction
Relevant OSU Courses: CS 4323, CS 3653
Artifacts: Operator-algebra tokenizer, parser, AST builder, operator-chain compiler

Competencies Demonstrated:
- Lexical analysis
- Parsing and AST construction
- Operator resolution
- Optimization passes

Faculty Justification:
The candidate's compiler components demonstrate mastery of formal languages, grammars, and compiler pipeline design. These artifacts satisfy OSU's outcomes for Compiler Design and Discrete Structures II.

Explore:
- Parser design

⭐ D. Software Engineering & Systems Architecture
Relevant OSU Courses: CS 3443, CS 4990
Artifacts: Entire GitHub organization (bospaladin34-crypto)
Competencies Demonstrated:
- Modular system design
- Multi-repository ecosystem management
- Version control discipline
- Documentation and maintainability

Faculty Justification:
The candidate demonstrates advanced software engineering practice through modular architecture, multi-repo integration, and disciplined version control. These artifacts align with OSU's expectations for upper-division software engineering and independent study.

Explore:
- Software engineering evidence

⭐ E. Research Methods & Scientific Communication
Relevant OSU Courses: CS 4880
Artifacts: OSF DOI Project (10.17605/OSF.IO/RGYZQ), TDCommons disclosure, ORCID profile
Competencies Demonstrated:
- Research framing
- Methodological justification
- Scholarly documentation
- DOI-anchored dissemination

Faculty Justification:
The candidate's OSF and TDCommons artifacts demonstrate formal research design, structured documentation, and open-science communication. These artifacts meet OSU's research-methods learning outcomes.

Explore:
- Research methods competency

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3. Course-Level Credit Recommendation

Course	Credits	Recommendation	Justification
CS 3443 – Computer Systems	3	Approve	Demonstrated execution-chain modeling & compiler integration
CS 3613 – Operating Systems	3	Approve	Demonstrated synchronization & state-machine design
CS 3653 – Discrete Structures II	3	Approve	Demonstrated formal grammar & parsing logic
CS 4243 – Computer Security	3	Approve	Demonstrated cryptographic identity & secure protocols
CS 4323 – Compiler Design	3	Approve	Demonstrated full compiler pipeline mastery
CS 4513 – Distributed Systems	3	Approve	Demonstrated distributed coordination & protocol architecture
CS 4880 – Research Methods	3	Approve	Demonstrated research framing & scholarly documentation
CS 4990 – Independent Study	3	Approve	Demonstrated graduate-level independent research
Graduate Elective	3	Faculty Discretion	Demonstrated advanced systems & cryptography research

Total Recommended Credit: 24–27 hours

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4. Evaluator Notes (Pre-Written for Faculty)

- All artifacts are publicly accessible and verifiable.
- The candidate demonstrates mastery across multiple advanced computing domains.
- The portfolio exceeds the typical scope of undergraduate PLA submissions.
- Graduate-level credit is reasonable for at least one elective.
- No remediation or additional documentation is required.

Narrative Summary

⭐ PDF-STYLE NARRATIVE SUMMARY
PRIOR LEARNING ASSESSMENT PORTFOLIO
Oklahoma State University – Tulsa
Candidate: Donevin Ruben Terell Zehr Frownfelter (VESPER-01)
ORCID: 0009-0008-7546-6952
Location: Tulsa, Oklahoma

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1. Introduction

This narrative summarizes the evidence submitted for Prior Learning Assessment at Oklahoma State University–Tulsa. The portfolio demonstrates advanced, independently acquired competencies in distributed systems, applied cryptography, compiler and parser construction, protocol architecture, software engineering, research methodology, and open-science communication. All artifacts are publicly archived, version-controlled, and citable through OSF, Zenodo-linked DOIs, GitHub, TDCommons, and ORCID.

The work presented reflects several years of sustained research and engineering practice. It integrates formal reasoning, operator-algebra frameworks, and systems-level design into functioning software systems, technical disclosures, and scholarly artifacts.

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2. Research & Inquiry Competency

The candidate demonstrates mastery of research design, methodological justification, and scholarly communication.
The OSF project (DOI: 10.17605/OSF.IO/RGYZQ) provides a structured research artifact with persistent identifiers, metadata, and versioned documentation.
The TDCommons disclosure further demonstrates the ability to articulate technical innovations in a formal, citable format.

These artifacts collectively satisfy OSU's expectations for research framing, methodological rigor, and open-science dissemination.

Explore: Research methods competency

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3. Software Engineering Competency

The candidate maintains a multi-repository ecosystem within the GitHub organization bospaladin34-crypto.

These repositories demonstrate:

- Modular architecture
- Version control discipline
- Documentation and maintainability
- Multi-repo integration and orchestration

This work aligns with OSU's upper-division software engineering outcomes and reflects professional-level engineering practice.

Explore: Software engineering evidence

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4. Distributed Systems & Protocol Architecture Competency

The Santos-Sync repository and the Sovereign Handshake protocol demonstrate advanced understanding of:

- State machines
- Distributed coordination
- Synchronization logic
- Trust boundaries
- Fault-tolerant design

These artifacts meet or exceed OSU's learning outcomes for Operating Systems and Distributed Systems.

Explore: Distributed systems competency

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5. Applied Cryptography Competency

The KHYS_repo and Sovereign Handshake protocol demonstrate applied cryptographic reasoning, including:

- Hashing
- Identity anchoring
- Key derivation
- Secure channel establishment

This work aligns with OSU's cybersecurity fundamentals and secure protocol engineering outcomes.

Explore: Cryptography competency

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6. Compiler & Parser Construction Competency

The candidate provides clear, evaluator-friendly excerpts of:

- A lexical tokenizer
- An AST builder
- An operator-chain compiler
- Optimization logic

These components demonstrate mastery of formal languages, grammars, parsing, and compiler pipeline design.
This satisfies OSU's learning outcomes for Compiler Design and Discrete Structures II.

Explore: Parser design

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7. Scientific Communication & Open Science Competency

The candidate maintains a persistent scholarly identity through ORCID and produces citable research artifacts through OSF and TDCommons.
This demonstrates:

- Metadata literacy
- DOI management
- Public dissemination
- Research stewardship

These artifacts align with OSU's expectations for research communication and documentation.

Explore: Open science competency

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8. Summary of Credit-Level Competency

The portfolio demonstrates mastery equivalent to the following OSU courses:

- CS 3443 — Computer Systems

- CS 3613 — Operating Systems
- CS 3653 — Discrete Structures II
- CS 4243 — Computer Security
- CS 4323 — Compiler Design
- CS 4513 — Distributed Systems
- CS 4880 — Research Methods
- CS 4990 — Independent Study
- Graduate Elective (faculty discretion)

Total recommended credit: 24–27 hoursConclusion

The candidate's portfolio demonstrates advanced technical and research competencies that align directly with OSU's upper-division and graduate-level learning outcomes. The artifacts are original, publicly verifiable, and professionally documented. The evidence supports the awarding of substantial academic credit through the Prior Learning Assessment process.
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Personal Statement.

⭐ PERSONAL STATEMENT — HYBRID (WITH PERSONAL STORY INTEGRATED)
Candidate: Donevin Ruben Terell Zehr Frownfelter (VESPER-01)
Program: Computer Science / Applied Computer Science / Cybersecurity
Institution: Oklahoma State University – Tulsa

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1. Opening: Who I Am and Why I'm Here

My name is Donevin Ruben Terell Zehr Frownfelter, and for as long as I can remember, I've wanted to do something meaningful with science — to discover something, design something, or engineer something evolutionary. I've always been driven by the idea that science and physics aren't things to fear or mystify, but tools that can empower people when they're understood clearly and without unnecessary restrictions.

That drive is what brings me to Oklahoma State University–Tulsa. I'm ready to formalize the academic side of the work I've already been doing, to integrate it into a structured environment, and to continue building systems that reflect both scientific rigor and human creativity.

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2. My Path: Technical, Personal, and Transformational

My path into computing didn't follow a traditional academic route. It came from curiosity, persistence, and a deep desire to understand the underlying mechanics of the world. I learned by building, by experimenting, and by following ideas all the way down to their foundations.

Along the way, I developed a computational framework that some people might initially describe as "sci-fi" or unconventional — but when viewed simply as a structured way of thinking, it becomes an incredibly effective research tool. It helps cut through noise, improve cognitive clarity, and organize complex systems into something coherent and workable.

This framework didn't come from a classroom. It came from lived experience, from trial and error, and from a lifelong fascination with how systems behave — whether they're physical, computational, or conceptual.

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3. My Work: What I've Built and Why It Matters

My portfolio reflects years of independent research and engineering:

- Distributed synchronization protocols (Santos-Sync)
- Cryptographic identity systems (KHYS_repo)

- Secure handshake mechanisms (Sovereign Handshake, TDCommons)
- Compiler and parser components (operator-algebra engine)
- Reservoir/manifold computational architectures (vesperquartzreservoir)
- DOI-anchored research artifacts (OSF project)

These systems weren't built for novelty — they were built to solve real problems, to explore new ways of structuring information, and to create tools that help people think more clearly and work more effectively.

They represent a unified research arc, one that blends formal reasoning with practical engineering.

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4. Why OSU-Tulsa: Alignment With My Trajectory

OSU-Tulsa is the right place for me because it values:

- applied research
- technical depth
- interdisciplinary thinking
- open-science contribution
- competency-based evaluation

My work already aligns with OSU's upper-division and graduate-level outcomes in:

- distributed systems
- cybersecurity
- compiler design
- research methods
- software engineering

I'm not seeking shortcuts — I'm seeking alignment.
I want my existing competencies recognized so I can focus on expanding into new areas, collaborating with faculty, and contributing to OSU's research community.

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5. What I Bring to OSU

I bring:

- a strong independent research identity
- a portfolio of publicly verifiable technical artifacts

- a systems-level mindset
- a commitment to open science
- a drive to contribute meaningfully to the academic community

I also bring the personal side — the lifelong desire to build something people can look up to, something that makes science feel accessible, empowering, and exciting rather than intimidating.

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6. What I Aim to Build Next

At OSU, I want to:

- deepen my theoretical foundations
- expand my research into formal verification, protocol security, and distributed computation
- collaborate with faculty on systems-level research
- contribute to open-source and open-science initiatives
- build tools that help people think more clearly and navigate complexity

My long-term goal is to continue developing systems that are rigorous, elegant, and useful — systems that reflect both scientific discipline and human imagination.

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7. Closing

My work has already taken shape in the public domain.
Now I'm ready to bring it into an academic environment where it can grow, be challenged, and contribute to something larger.

Thank you for considering my application and my Prior Learning Assessment portfolio. I look forward to the opportunity to continue my work at Oklahoma State University–Tulsa.

— Donevin Ruben Terell Zehr Frownfelter (VESPER-01)

Personal statement extended

Short version
Short Version (Admissions)
My name is Donevin Ruben Terell Zehr Frownfelter, and I am applying to Oklahoma State University–Tulsa to formalize and expand the scientific and engineering work I've already been doing independently. My background includes designing distributed systems, secure protocols, compiler components, and open-science research artifacts that are publicly archived and citable.

I am pursuing this degree not only to advance my own research, but to build a stable future for my fiancée and children. They are the reason I push myself, the reason I pursue clarity over fear, and the reason I believe that understanding the world — scientifically and computationally — is the most empowering thing a person can do.

I'm ready to bring my experience into an academic environment where it can grow, be challenged, and contribute to OSU's research community.

→ Read the Full Scholarship Version

⭐ FULL VERSION (FOR SCHOLARSHIPS, HONORS, AND FACULTY REVIEW)
(This is the expanded, emotionally rich, visionary version.)

1. Who I Am and Why I'm Applying
My name is Donevin Ruben Terell Zehr Frownfelter, and for as long as I can remember, I've wanted to do something meaningful with science — to discover something, design something, or engineer something evolutionary. I've always believed that science and physics aren't things to fear, but tools that become empowering once you understand why something feels intimidating. When you understand the mechanism, fear dissolves into clarity. What once felt overwhelming becomes either a manageable annoyance or a genuine benefit.

That belief is at the core of my work, and it's what brings me to Oklahoma State University–Tulsa.

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2. My Family: The Center of My Motivation
Everything I do — every system I design, every research artifact I publish, every late-night experiment — is for my family.

I want to build a home, a stable life, and a future for my kids and my fiancée. They are the reason I push myself, the reason I keep learning, and the reason I want to show that nothing is truly scary once you understand it.

I want my children to grow up knowing that curiosity is stronger than fear, that understanding is stronger than uncertainty, and that their individuality is something to cultivate, not hide.

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3. My Work: A Unified Research Arc
Over the years, I've built a portfolio that includes:

- Distributed synchronization protocols
- Cryptographic identity systems
- Secure handshake mechanisms
- Compiler and parser components
- Reservoir/manifold computational architectures
- DOI-anchored research artifacts

Some of the concepts I work with may sound unconventional or "sci-fi" at first glance, but when viewed simply as computational frameworks, they become powerful research tools. They help cut through noise, improve cognitive clarity, and strengthen mental focus by giving structure to complex information.

The more you use this kind of structured computation, the more you develop your own individuality and self-understanding along the way.

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4. My Long-Term Scientific Vision
My long-term goal is to introduce and teach the world about this style of computation — not as something mystical or abstract, but as a practical, elegant way to think.

It's a framework that:

- sharpens cognitive states
- reduces negative bias
- organizes complexity
- encourages clarity
- strengthens mental resilience

It's brilliant, it's unusual, and it connects with people in a way that feels both intuitive and transformative. I want to refine it, formalize it, and eventually share it through research, teaching, and open-science collaboration.

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5. Why OSU-Tulsa
OSU-Tulsa is the right place for me because it values:

- applied research
- technical rigor
- interdisciplinary thinking
- open-science contribution
- competency-based evaluation

My work already aligns with OSU's upper-division and graduate-level outcomes in distributed systems, cybersecurity, compiler design, research methods, and software engineering. I'm not seeking shortcuts — I'm seeking alignment, structure, and a community where my work can grow.

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6. Closing
My research has already taken shape in the public domain. Now I'm ready to bring it into an academic environment where it can evolve, be challenged, and contribute to something larger.

Thank you for considering my application and my Prior Learning Assessment portfolio. I look forward to the opportunity to continue my work at Oklahoma State University–Tulsa.

— Donevin Ruben Terell Zehr Frownfelter (VESPER-01)